

Epidemiological analysis of congenital heart defects: trends and outcomes from 2019 to 2023

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Abstract

Introduction: Congenital heart defects (CHD) are among the most common types of birth defects globally, affecting a significant number of children each year. In Kazakhstan, including the Turkestan region, there is growing concern over the increasing prevalence of CHD and the challenges associated with early detection and treatment. The COVID-19 pandemic has further complicated these issues by disrupting healthcare services, leading to delayed diagnoses and reduced access to specialized care. This study provides a comprehensive analysis of the prevalence and outcomes of CHD in the Turkestan region, focusing on recent trends and the impact of the pandemic.

This study aimed to analyze the incidence, prevalence, and outcomes of congenital heart defects among children in the Turkestan region of Kazakhstan between 2019 and 2023.

Methods: All patients diagnosed with congenital heart defects registered in the hospital during the specified period were included in this information-analytical study using ICD-10 codes Q20-Q28. Age trends, prevalence rates per 100,000 population, absolute mortality counts, and outcomes of hospital stay were calculated.

Results: The sample consisted of 5214 patients with congenital heart defects, of whom 890 were children under 1 year of age. During the study period, 2,374 surgical interventions were performed, and 173 in-hospital deaths were recorded.

Conclusions: The study showed that congenital heart disease in children in Turkestan region is characterised by a high incidence at the age of 1 to 3 years, a higher incidence in girls and an increase in cases during the COVID-19 pandemic. Most patients showed positive outcomes after hospitalisation and treatment, but mortality remained significant in cases of complex malformations.

Keywords: heart defects, congenital, epidemiology, child.

Introduction

Congenital heart defects (CHD) are structural abnormalities of the heart and major blood vessels that arise during embryonic or fetal development. These defects range from minor to life-threatening conditions. Depending on anatomical and hemodynamic characteristics, CHD are clinically classified into

different subtypes, each with a distinct spectrum of severity and symptoms.

The prevalence, mortality and morbidity associated with CHD remain significant, especially in developing countries. In the 2019 GBD study [1], a total of 3.12 million (95% CI: 2.40-4.11 million) children were born with congenital heart anomalies,

representing a prevalence of 2,305.2 per 100,000 live births (95% CI: 1,772.9-3,039.2 per 100,000 live births). Important data were also collected on rates associated with congenital cardiovascular anomalies (CHA) expressed in terms of disability-adjusted life years (DALYs), years lost to premature death (YLLs), and years lived with disability (YLDs). The overall DALYs for cardiovascular disease was 241.6 per 100,000 (95% UI: 196.1-292.7), indicating a significant disease burden. YLLs were 234.0 per 100,000 (95% UI: 189.8-285.7), indicating a high loss of life years due to premature death. In contrast, YLDs were 7.6 per 100,000 (95% UI: 3.7-12.7), showing that although cardiovascular disease causes significant loss of life years, the number of years spent with disability is relatively low. In addition, CHA was the leading cause of 217,000 deaths (95% UI: 177,000 to 262,000 deaths), of which 150,000 deaths (95% UI: 120,000 to 184,000 deaths) occurred in infants under 1 year of age.

The challenges of pediatric and adult CHD care vary from country to country. In Asian countries, there is a different picture of CHD care. In some developed countries, such as Japan and South Korea, the number of adult CHD patients exceeds the number of children with CHD. However, in low- and middle-income countries, diagnosis and treatment in early childhood may be inadequate, resulting in a higher number of school-aged children with untreated CHD. For example, in South Korea, the proportion of patients with Adult congenital heart disease (ACHD) among outpatients increased from 11% in 2005 to 32% in 2017 [2].

The care of children with CHD is resource-intensive and expensive. CHD hospitalizations alone cost about \$5.6 billion per year, which is 15% of the cost of all pediatric inpatient care. This is especially important when considering that CHD hospitalizations account for only 3.7% of all pediatric hospitalizations [3, 4]. In addition, more than 40,000 surgical and catheter interventions are performed annually in 120 pediatric cardiology programs [5].

In Kazakhstan, epidemiological data on hospital admissions with congenital heart disease (CHD), especially in paediatric cohorts, are under-researched compared with data from high-income and developing countries. The first national study using a single electronic health system showed that 68,371 patients with CHD were registered between 2019 and 2021, of whom 59.6% were diagnosed in the first year of life. The mortality rate for multiple CHD was 11.5% and 7.2% for a single type, highlighting the need for further study [6].

The relevance of studying the problem of CHD is associated not only with the high incidence of disability and mortality, especially among children, but also with the need to improve medical care. CHDs require significant health care resources for diagnosis, treatment and surgical interventions, which remain complex and expensive. In addition, the timely detection and treatment of CHD contributes to reducing the economic burden on the family and society associated with prolonged treatment and rehabilitation.

This study aimed to analyze the incidence, prevalence, and outcomes of congenital heart defects among children in the Turkestan region of Kazakhstan between 2019 and 2023.

Methods

Sampling

This study is informational and analytical in nature. To assess the epidemiological characteristics of congenital heart

defects (CHD) in the Turkestan region of the Republic of Kazakhstan, statistical data of hospitalized patients from the Regional Clinical Children's Hospital were analyzed for the period from January 1, 2019, to December 31, 2023. Patients with CHD sought medical care and were referred to this hospital from 14 districts and 4 regionally subordinated cities for treatment [7]. The statistical data of patients, containing socio-demographic and clinical information, were aggregated using the International Classification of Diseases, 10th Revision (ICD-10). CHD cases were identified using ICD-10 codes Q20-Q28. The diagnosis was either previously established or made for the first time during hospitalization.

Exposure and Covariates

Patient records extracted from statistical data contained the following information: date of birth, gender, place of residence and healthcare attachment, type of hospitalization (emergency/planned), ICD-10 codes for the primary diagnosis, admission and discharge dates, length of hospital stay (in days), type of surgical intervention, and an anonymous population registration number (RPN).

Inclusion criteria: All patients diagnosed with CHD according to ICD-10, residing in and attached to primary healthcare facilities (PHC) in the Turkestan region. Exclusion criteria: Patients who were not residents of the Turkestan region or were attached to PHC facilities in other regions of Kazakhstan, as well as duplicate cases and repeated hospitalizations of the same patients within a single year. Age at the time of hospitalization was categorized as follows: under 1 year, 1–3 years, 4–7 years, 8–12 years, 13–15 years, and 16–18 years. Residential location was classified as either "urban" or "rural" based on the patient's PHC attachment and place of residence, distinguishing between those living in cities and those in areas outside large and small cities.

Statistical Analysis

All statistical analyses were performed using IBM SPSS, version 27.0. Qualitative variables were expressed as simple frequencies (absolute numbers) and relative frequencies (percentages). Quantitative variables were presented as means, medians, and standard deviations, with the range (extreme values) specified.

Incidence and prevalence rates were calculated by dividing absolute case numbers by the population of the Turkestan region for the corresponding year and multiplying by 100,000. All-cause mortality was presented as absolute numbers, as mortality rates per 100,000 were not computed. Population data were obtained from the Taldau Statistics information-analytical system [8].

Ethical Considerations

This study was conducted as part of a doctoral dissertation and adhered to both international and local ethical guidelines and regulations. It received approval from the local bioethics committee of the South Kazakhstan Medical Academy (Protocol №3), Shymkent, Kazakhstan. All data were anonymized prior to analysis, in accordance with the principles of the Declaration of Helsinki.

Results

General Characteristics of the Sample. The final study sample included 5,214 hospitalizations of patients diagnosed with congenital heart defects (CHD), of which 2,884 cases were

female and 2,330 were male. The average age of patients at the time of hospitalization was 3,26 (0,03-11,23) months. Among the hospitalized patients, 890 cases were recorded in children under 1 year of age, accounting for 17.1% of the total.

Among all hospitalizations, 79.9% were planned admissions, while emergency hospitalizations accounted for 20.1%. Notably, planned hospitalizations exceeded emergency cases in all study years (Figure 1). Most patients were from rural areas, totaling 2,877 individuals. They were predominantly admitted for planned hospitalizations (78.1%), with emergency hospitalizations making up 21.9%. In contrast, the urban patient population consisted of 2,337 individuals.

The number of fatal outcomes among hospitalized cases was 173. The majority of patients exhibited favorable outcomes, with clinical improvement or complete recovery observed in most cases. Mortality rates remained low throughout the study period (Figure 2).

Peripheral arteriovenous malformation (Q27.3) was the most common condition, accounting for 1,563 (29.9%) cases among CHD patients (Table 1). During the study period, a total

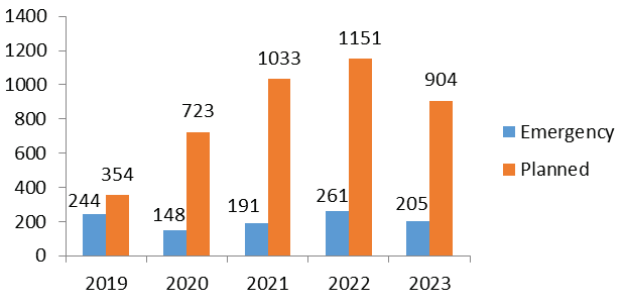


Figure 1 – Ratio of planned and emergency hospitalizations of CHD patients (2019–2023)

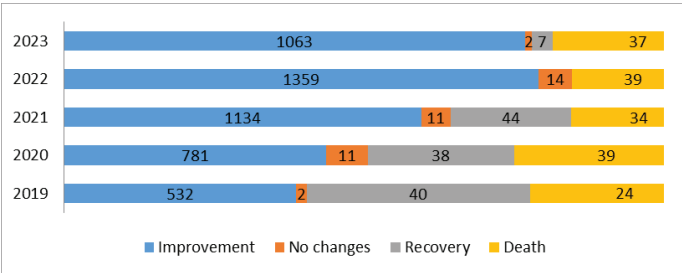


Figure 2 – Hospitalization outcomes of patients with CHD (2019–2023)

Table 1

Most common congenital heart defects and data on surgical interventions and fatal outcomes (2019–2023).

Diagnoses with the corresponding ICD-10 codes	Number of patients with the given diagnosis as a single malformation, N (%)	Number of patients who underwent surgery for this diagnosis, N (%)	Number of patients who died from this diagnosis, N (%)
Q 27.3 Peripheral arteriovenous malformation	1563 (29, 9%)	1294 (54, 5%)	2 (1, 2%)
Q 21.0 Ventricular septal defect	1155 (22,1%)	283 (11,9%)	15 (8,7%)
Q 21.1 Atrial septal defect	522 (10%)	240 (10,1%)	3 (1,7%)
Q 21.3 Tetralogy of Fallot	287 (5,5 %)	73 (3%)	16 (92%)
Q 25.0 Patent ductus arteriosus	271 (5,1%)	182 (7,7%)	3 (1,7%)

of 2,374 surgical interventions were performed for the treatment, correction, and diagnosis of congenital heart defects.

Trends in morbidity and mortality

As illustrated in Figure 3, the distribution of congenital heart defect (CHD) cases by age group from 2019 to 2023 reveals a consistent pattern, with the highest incidence recorded among children aged 1 to 3 years. In 2019, this group accounted for 212 cases, which increased slightly to 227 in 2020 and peaked at 245 in 2021. Although there was a minor decline in subsequent years—230 cases in 2022 and 218 in 2023—this age group remained the most affected throughout the study period. Infants under the age of one year also demonstrated a high burden of CHD, with case numbers ranging from 150 to 180 annually. In contrast, children aged 4 to 6 years and 7 to 14 years showed progressively lower incidence rates, while adolescents aged 15 to 17 years and adults (18 years and older) accounted for the fewest cases across all years.

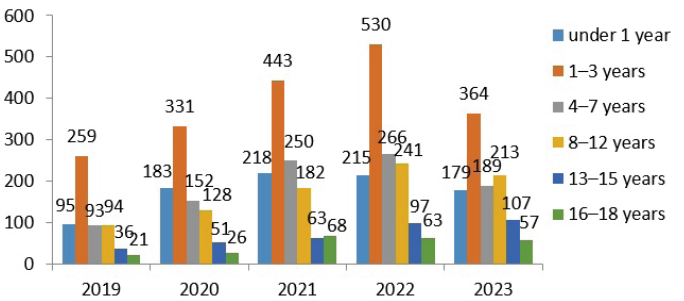


Figure 3 – Age-related trends in the incidence of congenital heart defects, presented in absolute numbers

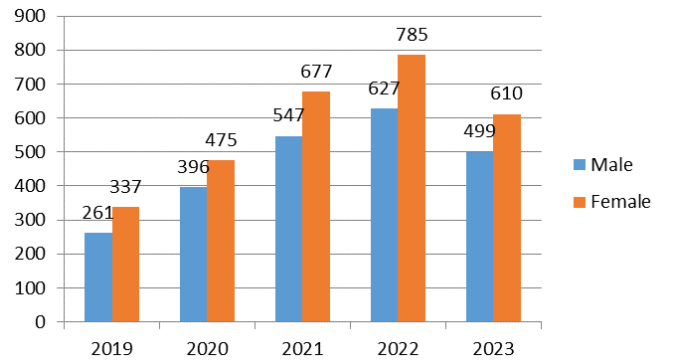


Figure 4 – Absolute number of hospitalized patients diagnosed with CHD, stratified by sex and year of admission (2019–2023)

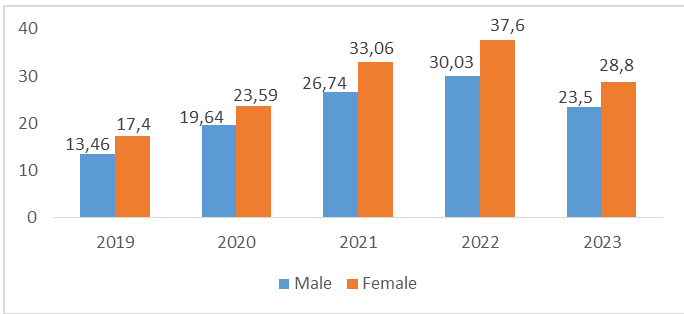


Figure 5 – Hospitalization rate of patients diagnosed with CHD per 100,000 population, stratified by sex and year of admission (2019–2023)

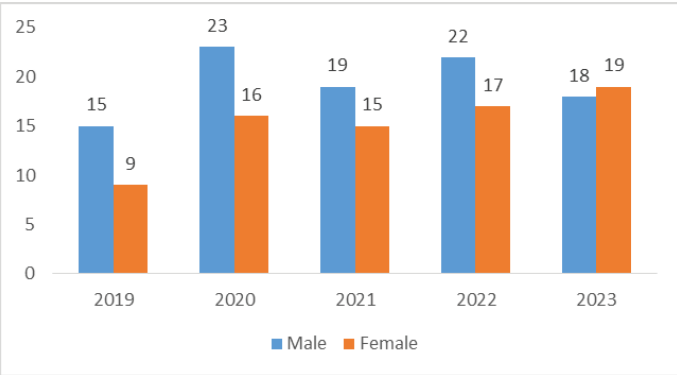


Figure 6 – Mortality rate of hospitalized patients diagnosed with CHD, stratified by sex and year of admission (2019–2023), presented in absolute numbers

Figure 5 demonstrates that throughout the study period, the incidence of congenital heart defects was higher among female patients. The incidence showed a steady increase from the beginning of the study until 2022, rising from 13.46 to 30.03 cases per 100,000 population among men and from 17.4 to 37.6 cases among women. In 2023, a decline in incidence was noted, reaching 23.5 cases per 100,000 population among men and 28.8 cases among women.

Mortality among hospitalized patients diagnosed with CHD was higher in men (Figure 6). In 2020, it peaked at 23 cases, while in women, the mortality rate gradually increased from 9 cases in 2019 to 19 cases in 2023.

Among the analyzed cases, four types of CHDs stand out due to their rarity, each occurring only once over the five-year period: Congenital septal defect between the aorta and pulmonary artery (Q21.4), Unspecified congenital anomaly of the peripheral vascular system (Q27.9), Arteriovenous developmental anomaly of precerebral vessels (Q28.0), and Unspecified congenital anomaly of the aortic and mitral valves (Q23.9). Their rare occurrence highlights the need for careful monitoring and timely diagnosis of such anomalies (Table 1).

Discussion

During the study period, children under one year of age accounted for 17.1% of the total number of hospitalized patients with congenital heart defects (CHD), while 82.9% of hospitalized children were older than one year. According to large-scale national studies, 40.4% of CHD diagnoses are made after the first year of life [6].

The high proportion of hospitalized patients over one year old with CHD may be explained by the potentially high prevalence of undiagnosed cases during the perinatal period. However, other researchers argue that prenatal diagnostics is an effective method for detecting complex and combined CHDs in the fetus, as well as for determining management strategies for newborns from high-risk groups [9–10]. Many patients with congenital heart defects who require surgery within the first six months of life remain undiagnosed before birth [11].

To address this challenge and improve early detection, screening methods including pulse oximetry are used. Furthermore, the European Society of Cardiology (ESC) has developed recommendations for the use of NT-proBNP and BNP as reliable biomarkers for the diagnosis and prognosis of heart failure in newborns. NT-proBNP is characterized by greater stability and a longer half-life, is synthesized in response

to increased pressure in the heart chambers, correlates with the severity of left ventricular dysfunction, and helps differentiate cardiac from respiratory causes of dyspnea [12–13].

Supporting these recommendations, a study by Azizova [14] demonstrated that NT-proBNP levels in critically ill infants with CHD were on average 10 times higher than those in healthy newborns ($P < 0.001$).

Therefore, effective prenatal CHD screening should be a public health priority, as early detection of these defects in childhood helps prevent initial diagnosis in adulthood and the associated complications [15].

According to our study results, adolescents aged 16–18 years show the lowest incidence rates, ranging from 21 to 68 cases, indicating a consistently low morbidity level in this category throughout the study period. This may be related to the transition of patients in this age group from pediatric to adult healthcare services [16–20].

The sharp increase in CHD-related hospitalizations observed in 2021–2022 may be attributed to the effects of the COVID-19 pandemic, including its impact on the cardiovascular system, the presence of comorbid conditions, and increased detection rates. Global studies have confirmed that COVID-19 can cause various types of cardiovascular damage in previously healthy infected individuals. Moreover, it has been established that the generally milder course of COVID-19 in children does not apply to cases involving congenital heart defects (CHD) [21–23].

In our study, the most common types of CHD among hospitalized patients were PAM, VSD, ASD, TOF, and PDA. Our findings align with the results of Marmech’s study [24], in which 68% of 7,926 hospitalized patients had left-to-right shunt pathology. The most frequent conditions were ventricular septal defect (VSD) and atrial septal defect (ASD). Ductal-related conditions, including Tetralogy of Fallot (TOF), aortic coarctation, and pulmonary atresia with VSD, were observed in 16% of cases.

The majority of patients admitted for planned hospitalization were residents of rural areas. This may be due to the fact that all major cardiology and cardiac surgery hospitals specializing in CHD treatment are located in cities of national significance. As a result, patients from rural areas are referred to these facilities for more in-depth diagnostics, surgical treatment, and rehabilitation care.

This study has several limitations. Firstly, it is based on a relatively small sample size and incomplete data, which may have been affected by human factors during collection and processing. Secondly, there was no access to information regarding patients’ socioeconomic status, education level, or quality of healthcare services, which are important factors in understanding CHD-related outcomes.

In the present study, mortality among hospitalized patients with congenital heart defects (CHDs) per 100,000 population in the Turkestan region for the period 2019–2023 was not calculated due to the lack of complete and reliable data on the number of deaths, as well as the inability to identify deaths specifically caused by CHDs.

Additionally, our prevalence estimates are limited to children aged 0–18 years and may differ from studies focusing only on infants under one year of age. Data from previous years were not included due to unavailability, which may have led to an underestimation of the true prevalence.

Conclusions

This study revealed that CHD in children in the Turkestan region are characterized by a high incidence in the 1–3-year-old age group, a higher frequency among girls, and an increase in cases during the COVID-19 pandemic. The most common types of CHD were arteriovenous malformations, ventricular and atrial septal defects, Tetralogy of Fallot, and patent ductus arteriosus.

The majority of patients demonstrated positive outcomes following hospitalization and treatment; however, mortality remained significant in cases of complex defects. Limited effectiveness of prenatal diagnosis and restricted access to specialized care in rural areas remain key challenges.

Strengthening early screening programs and ensuring equitable access to pediatric cardiac services, especially in rural settings, are essential steps in reducing the burden of CHDs. Further multicenter and longitudinal studies are required to validate these findings and guide public health policies.

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